

Discrete Structure

BCA — Semester 2 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Explain direct proof, indirect proof, and proof by contradiction. Use direct proof to show that "If n is an odd integer, then n^3 is an odd integer". Also use indirect proof to show that "If n is an integer and n^3 then n is odd".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is linear nonhomogeneous recurrence relation of degree k with constant coefficients? Find all the solutions of the recurrence relation $a_{n+k} = a_n + n$. Also find the solution of the relation with initial condition $a_1 = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define spanning tree and minimum spanning tree with suitable example. Use Kruskal's algorithms to find minimum spanning tree in the given graph.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is tautology? Show $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define cartesian product. Find A^3 for the set $A = \{a, b, c\}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How can you represent relations using matrices? Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What matrix representing R if $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Use mathematical induction to show that the sum of first n positive integers is $n(n+1)/2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is congruent modulo? Determine whether 20 is congruent to 8 modulo 6 and 25 is congruent to 17 modulo 5.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain trial division with example? Using trial division, show that 101 is prime.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain product rule. How many strings are there of four lowercase letters that have the letter x in them?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is graph? Explain simple graph and pseudograph with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is Euler path? Compare it with Hamilton path.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.